



PlantDoctor

Offline AI-Powered Diagnostics for Industrial Systems

AI/ML Hackathon Entry

● The Problem: The Impossible Debugging Challenge

Why This Matters

Manufacturing plants operate offline by necessity:

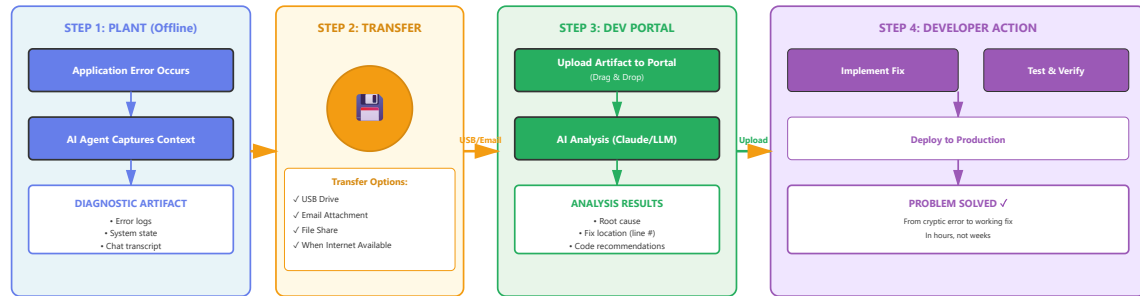
- X Safety compliance requires offline systems (no external network dependency)
- X When critical applications crash, operators see cryptic errors
- X Developers can't reproduce the problem locally (missing context)

Result: 2+ weeks debugging = \$100K-\$500K lost per incident

Real Impact: Every manufacturing company faces this. We experienced it personally while working in plant automation.



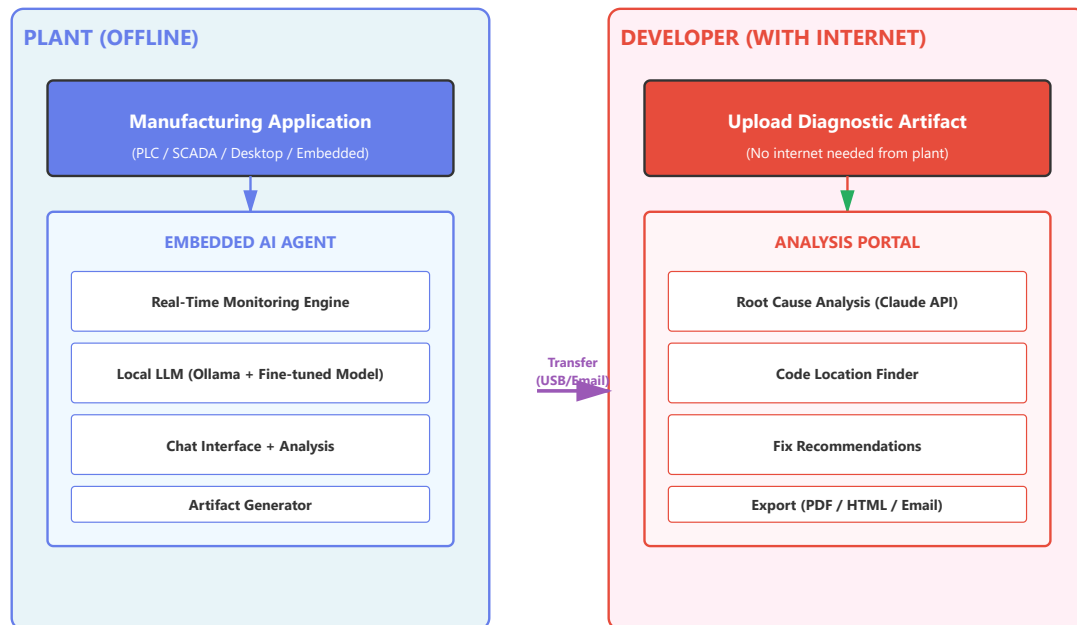
How It Works: Offline Plant → Developer in 4 Steps



Complete Workflow: Error occurs offline → AI captures → Transfer via USB → Developer analyzes → Problem fixed



System Architecture: How It Works



Key Innovation: Everything on the left runs OFFLINE. Zero internet dependency. The LLM is fine-tuned on your application's codebase.

✓ The Solution: AI That Understands Your Code

How PlantDoctor Works

- 1. Real-Time Monitoring:** Agent monitors logs, system state, metrics 24/7
- 2. Error Detection:** Instantly detects when something goes wrong
- 3. Context Capture:** Records complete state (CPU, memory, database, user actions)
- 4. AI Analysis:** LLM analyzes error against code knowledge base
- 5. Operator Chat:** "What were you doing?" → AI refines diagnosis
- 6. Artifact Export:** Complete diagnostic ready for developers
- 7. Developer Fix:** Receives actionable intelligence (code location, line number)

★ Key Features



100% Offline

Zero internet dependency. Works where plants actually are.



Context-Aware

LLM trained on YOUR code, understands YOUR system.



Real-Time



Operator Chat

<500ms analysis. Instant diagnostics, no cloud latency.

Simple interface. No technical knowledge needed.



Smart Export

Complete artifacts with context, logs, diagrams, transcripts.



Developer Ready

Code locations, line numbers, actionable fix suggestions.



Why This Is Innovative

The Novel Approach

- ✓ **Offline-First:** Most AI solutions are cloud-first. We're offline-first because that's where the problem exists.
- ✓ **Application-Specific:** LLM is fine-tuned on YOUR code, not generic patterns.
- ✓ **Context Preservation:** Captures complete system state, not just error message.
- ✓ **Human-AI Collaboration:** Operator + AI working together beats either alone.
- ✓ **Production-Grade:** Runs embedded in real applications, not experimental.



Expected Impact Metrics

97%

MTTR Improvement
(14 days → 4 hours)

85%+

Diagnostic Accuracy
With operator input

<500ms

Analysis Latency
No cloud wait

100%

Offline Capable
Zero internet needed

Flexible Technology Stack

Choose based on your team's expertise:

LLM Inference

- Ollama
- LocalAI
- vLLM

Models: LLaMA 2, Mistral, CodeLLaMA

Backend

- Python + FastAPI
- Node.js + Express
- Go + Echo

Pick what you know

Frontend

- React + Vite
- Vue.js
- Angular/Svelte

Your preference

Storage

- SQLite (simple)
- JSON files
- In-memory cache

Lightweight & fast



Let's Build Something Great

This solves a real problem that every manufacturing company faces.

From "I can't reproduce it" to "Here's the exact fix" in 4 hours instead of 2 weeks.

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Offline AI-Powered Diagnostics for Industrial Systems

- 📌 Problem: Manufacturing plants operate offline, making debugging impossible
- 💡 Solution: Embedded AI agent with locally-deployed LLM
- 🎯 Result: 97% MTTR improvement + actionable developer insights